

Education

2023–27 **University of Maryland, College Park**
(exp. May 27) B.S. in Computer Science (Honors), B.A. in Linguistics
Minor in Korean Studies
Advisors: [Rachel Rudinger](#), [Jordan Boyd-Graber](#), & [Mohit Iyyer](#)
Coursework: Natural Language Processing (G), Commonsense Reasoning (G), Mechanistic Interpretability (G)*, Machine Learning, Data Science, Parallel Computing, Computer Systems, Syntax, Semantics*, Phonetics, Psycholinguistics (* in progress)

Research Experience

- 2024– **Undergraduate Researcher**, CLIP Lab, University of Maryland, College Park
Advisors: [Rachel Rudinger](#), [Jordan Boyd-Graber](#), & [Mohit Iyyer](#)
AI agent feedback for long-form text generation
Calibrating human reliance and susceptibility to VQA model explanations
Artifacts and distractor quality in human-written vs. LLM-generated MCQs
Adversarial benchmarking of VLMs on out-of-context video misinformation
Causal interpretability of syntactic learning in data-constrained LMs
- 2024– **Visiting Researcher**, ACL2 Lab & NSF, The University of Texas at Arlington
Advisor: [Kenny Zhu](#)
Unsupervised techniques for feline language discovery and categorization
[AniVoice-cat](#): 26,000+ annotated cat vocalizations and 57 discovered phones
- 2025–26 **Member of Technical Staff**, Learn Prompting (San Francisco, CA)
Ran HackAPrompt, the world’s largest AI red-teaming hackathon
Robustness of AI safety judges to universal and multimodal CBRNE jailbreaks
- 2023–24 **Researcher**, FIRE Sustainability Analytics Lab, University of Maryland
Advisor: [Thanicha Ruangmas](#)
Climate policy frameworks based on automated OCR of 50,000+ 10-K filings
- 2020–23 **Researcher**, Reinforcement Learning at Brown Group, Brown University
Advisor: [Michael Littman](#)
Human-readable reward interfaces letting non-experts specify RL tasks

Publications

[Google Scholar](#) * – equal contribution

Under Submission

- P1. **Atrey Desai**, Nishant Balepur & Rachel Rudinger. *Quick, Create a Distractor! Evaluating LLM Distractors for Multiple-Choice Benchmarks*. Under review at ACL Rolling Review (ARR).
- P2. Tuan Dang*, **Atrey Desai***, Hridayesh Lekhak*, Tirza Panunto*, Lindsay Pike*, Theron Wang* & Kenny Zhu*. *A Preview of Computational Animal Linguistics*. Under review at ACM Computing Surveys.

Peer-reviewed Proceedings

- C1. **Atrey Desai** & Sathvik Nair. *Filling in the Mechanisms: How do LMs Learn Filler-Gap Dependencies under Developmental Constraints?* ACL 2026 (Findings). ★ **Oral, TLS 2026**
- C2. Nishant Balepur, Bhavya Rajasekaran, Jane Oh, Michael Xie, **Atrey Desai**, ..., Jordan Boyd-Graber. *BenchMarker: An Education-Inspired Toolkit for Highlighting Flaws in Multiple-Choice Benchmarks*. ACL 2026.
- C3. Nishant Balepur, **Atrey Desai** & Rachel Rudinger. *Test-Time Reasoners Are Strategic Multiple-Choice Test-Takers*. ACL 2026.
- C4. **Atrey Desai**, Nishant Balepur & Rachel Rudinger. *Language Models Generate Multiple-Choice Questions with Artifacts*. MASC-SLL 2025.
- C5. Chace Hayhurst, Hyojae Park, **Atrey Desai**, Suheidy De Los Santos & Michael Littman. *Reinforcement Learning As End-User Trigger-Action Programming*. IML Workshop @ AAAI 2022; RLDM 2022.

Selected Awards & Honors

- 2026 CRA NSF CISE Student Funding Program (\$9,000)
- 2026 ACL D&I Subsidy (\$750)
- 2026 Department of Computer Science Professional Development Fund (\$500)
- 2025 SPIRE Research Grant (\$3,000)
- 2024 Omicron Delta Kappa Top 10 Freshman
- 2023–26 CMSC & ARHU Dean’s List
- 2023 UMD President’s Scholarship (\$50,000)
- 2023 NMSC National Merit Scholarship (\$4,000)
- 2023 Catherine Yang Scholarship (\$1,000)

Talks

- 2026 **Filler-Gap Dependencies under Developmental Constraints in LMs**
Texas Linguistics Society (Feb 2026)
- 2026 **Filler-Gap Dependencies under Developmental Constraints in LMs**
UMD Computational Cognitive Science Reading Group (Feb 2026)
- 2024 **Adaptor Grammars and Neural Networks for Feline Lexical Discovery**
University of Maryland (Nov 2024)
- 2024 **Adaptor Grammars and Neural Networks for Feline Lexical Discovery**
University of Texas at Arlington (Jul 2024)

Service & Leadership

- 2026– **Organizer**, [QANTA/EMM-QA](#) (2026)
- 2025– **Reviewer**, *ACL/ACL Rolling Review (ARR; Jan 2026, May 2026), ICML EMM-QA, NeurIPS IAEval
- 2025– **Boardmember**, [University Research Advisory Board](#)

- 2025–26 **Vice Chair**, [Computer Science Department Council](#)
Appointed by faculty and student body; represent 4,200+ CS undergraduates.
- 2024–26 **Senior Member**, [FIRE Student Leadership Council](#)
Represented 1000+ peers, ran events & workshops, and reformed class curricula.
- 2024– **University Ambassador** ([Department](#) & [College](#))

Mentoring

- 2026 **Summer Research:** Nathan Nguyen, Nghia Truong, Yasmine Tajeddin
- 2025 **Office of Undergraduate Research:** Juan Cortés, Kemisola Benson, Vivian Akpala
- 2024 **Technica:** Savya Miriyala, Tanya Grover, Jessica Ononye, Nakshatra Hiray
- 2020–22 **Robotics Curriculum:** designed and led for [MSET](#)

Skills

- Programming Python, Java, JavaScript/HTML/CSS, R, MATLAB
- Frameworks Huggingface Transformers/Datasets, NLTK, PyTorch, Selenium, BeautifulSoup
- Tools Git, Docker, GCP, Google Vertex AI, VS Code
- Languages English (Native), Gujarati (Native), Spanish (Intermediate), Korean (Beginner)
- Forecasting **Top-3 global forecaster** on Manifold Markets (Mar–July 2026); Top 10 all-time on AI topics